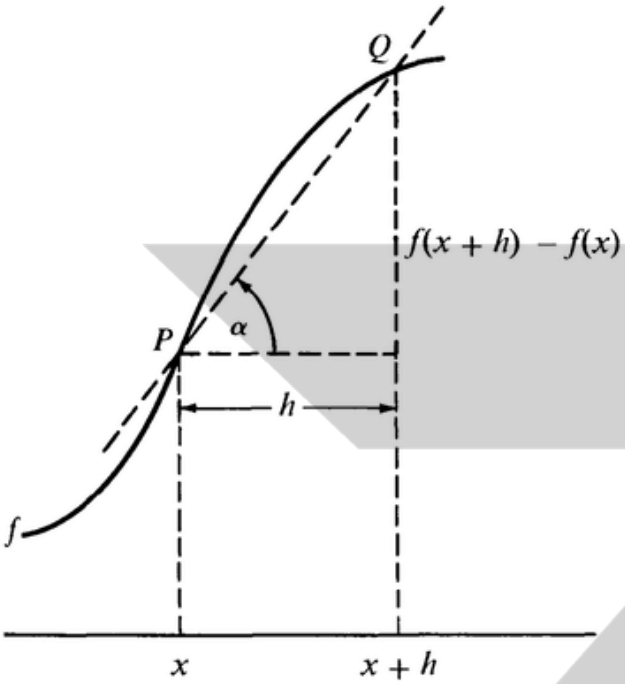
1. Differentiation

1.1. Introduction

- Differentiation helps us find the **gradient of the tangent**, a line which touches the curve at a **single point**.



Using the formula for finding the gradient, we get

$$m = \frac{f(x+h) - f(x)}{h}$$

Where h is a small number that is greater than 0.

- As h becomes smaller and approaches 0 (but never reaches it), we get the gradient of a tangent to a point on the curve.

From this, we can deduce

$$f\prime(x) = tan(\theta) = m$$

- Where $f\prime(x)$ represents the **first derivative**, or the result obtained by differentiating $f(x)$ once.

Notation

It is denoted by many notations:

$$\frac{dy}{dx} = f\prime(x) = y\prime$$

- Where $\frac{dy}{dx}$ implies a function y has been differentiated with respect to the variable x.
  - It can also be written as $\frac{d}{dx}y$, where y is the function being differentiated.

Basics of Differentiation

Power rule

$$\frac{d}{dx}x^n = n \times x^{n-1} \quad \text{[For any real number } n \neq 0\text{]}$$

Sum and Difference rule

$$\frac{d}{dx}f(x) \pm g(x) = \frac{d}{dx}f(x) \pm \frac{d}{dx}g(x)$$

Scalar multiple rule

$$\frac{d}{dx}a \times f(x) = a \times \frac{d}{dx}f(x) \quad \text{[Where } a \text{ is a constant number]}$$

Constant rule

$$\frac{d}{dx}a = 0 \quad \text{[Where } a \text{ is a constant number]}$$

1.2. Chain rule

- The chain rule helps us differentiate a **composite function**.

Consider $f(\,g(x)\,)$. If we let $u = g(x)$ and $y = f(u)$, and then differentiate y, we get

$$\frac{dy}{dx} = \frac{du}{dx} \times \frac{dy}{du}$$

So we can define the derivative of $y = f(\,g(x)\,)$

$$\frac{d}{dx}f(\,g(x)\,) = f\prime(\,g(x)\,) \times g\prime(x)$$

It may also be thought of like this

$$\underbrace{f'}_{\text{derivative of outside function}} \quad \underbrace{(g(x))}_{\text{inside function left alone}} \quad \underbrace{g'(x)}_{\text{times derivative of inside function}}$$

Applications of chain rule

- The chain rule can be applied to differentiate functions of the type $[f(x)]^n$

Using the chain rule and power rule, we can differentiate $y = [f(x)]^n$ to get

$$\frac{d}{dx}[f(x)]^n = n \times [f(x)]^{n-1} \times f\prime(x)$$

Another application is seen while differentiating $y = (ax+b)^n$

$$\frac{d}{dx}(ax+b)^n = n \times (ax+b)^{n-1} \times \frac{d}{dx}ax+b$$

$$\implies \frac{d}{dx}(ax+b)^n = n \times (ax+b)^{n-1} \times a$$

- This is valid for all real numbers $n \neq 0$

Example

Differentiate $y = (2x+5)^5$.

Answer

Using the chain rule

$$\frac{d}{dx}(2x+5)^5 = 5 \times (2x+5)^{5-1} \times \frac{d}{dx}2x+5$$

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$$\implies \frac{d}{dx}(2x+5)^5 = 5 \times (2x+5)^4 \times 2$$

$$\implies \frac{d}{dx}(2x+5)^5 = 10 \times (2x+5)^4$$

1.3. Gradient function

- The first derivative is also called the gradient function. It gives us the gradient of the tangent to the curve at a specific point.

A general equation of a tangent can be found using

$$y = f\prime(x)x + c$$

To find the gradient of the tangent at the point (a, b), input x = a into the first derivative.

$$f\prime(a)$$

Using this, we can re-write the **equation of the tangent** to the curve at the point (a, b) as

$$y = f\prime(a)x + c$$

Equation of perpendicular

- To find the equation of a perpendicular to a point, we must first find the equation of the gradient of the tangent at that point on the curve.

We can consider the gradient of the tangent at (x, y) to be $f\prime(x)$

Using the fact that the gradients of a perpendicular and a tangent follow the relation:

$$m_1 \times m_2 = -1$$

We can re-write the gradient of the perpendicular as

$$f\prime(x) \times m_2 = -1 \implies m_2 = -\frac{1}{f\prime(x)}$$

Example 1

Find the gradient of the tangent to the curve $y = 5x^2 + 3x$ at the point $x = 2$.

Answer

We first start by finding the gradient function, or the first derivative.

$$\frac{d}{dx}5x^2 + 3x = 10x + 3$$

Now we can input $x = 2$ into $f\prime(x)$

$$f\prime(2) = 10(2) + 3 = 23$$

So the gradient of the tangent to the curve at $x = 2$ is 23.

Example 2

A curve is described by the equation

$$f(x) = \frac{1}{\sqrt{x}} \text{ for } x > 0$$

The point A is on the curve such that the normal to the curve at A also passes through the origin. Find the coordinates of the point A

Answer

- With the given information, the point A can be considered the point of intersection of the curve $f(x)$ and the perpendicular to the curve at that point.

To find the gradient of the normal, we first find the general gradient of a tangent to the curve by differentiating the curve once.

$$f\prime(x) = \frac{d}{dx}x^{-\frac{1}{2}} = -\frac{1}{2x^{\frac{3}{2}}}$$

Now we can use the relationship between the gradients of a perpendicular and a tangent

$$-\frac{1}{2x^{\frac{3}{2}}} \times m_2 = -1 \implies m_2 = -\frac{1}{-\frac{1}{2x^{\frac{3}{2}}}}$$

So the gradient of the normal is

$$m_2 = 2x^{\frac{3}{2}}$$

Using this gradient, and the y-intercept as 0 (due to the question stating that the normal passes through the origin), we get the equation of the normal as

$$y = mx + c \implies y = 2x^{\frac{3}{2}} \times x + 0$$

$$y = 2x^{\frac{5}{2}}$$

Equating $f(x)$ and the equation of the perpendicular will give us the x coordinate of A.

$$x^{-\frac{1}{2}} = 2x^{\frac{5}{2}} \implies x = 2^{-\frac{1}{3}}$$

Inputting this x-value into any one equation will give us the corresponding y-coordinate of A.

$$y = \left(2^{-\frac{1}{3}}\right)^{-\frac{1}{2}} \implies y = 2^{\frac{1}{6}}$$

So the coordinates of A are

$$\left(2^{-\frac{1}{3}}, 2^{\frac{1}{6}}\right)$$

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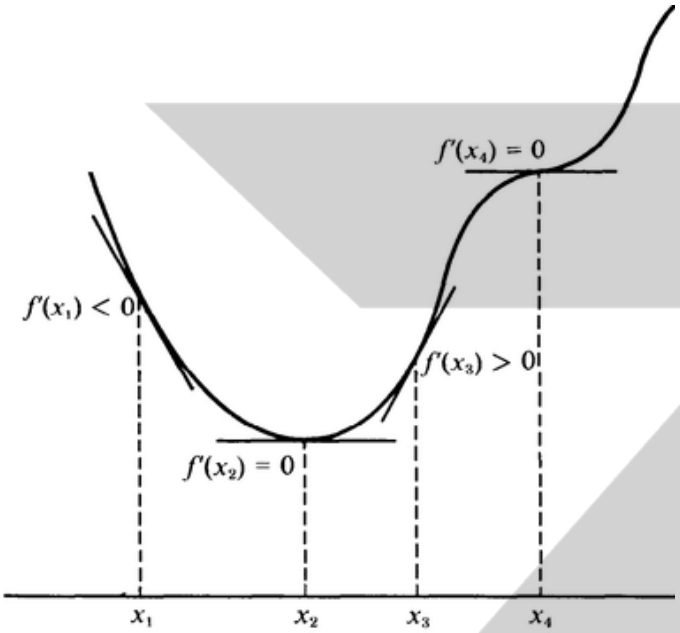
1.4. Rate of change

- The rate of change of a function can be connected to the rate of change of its gradient.
  - A **positive gradient** means the function has an **increasing** y-value as the x-value increases. This is called an **increasing function**.
  - A **negative gradient** means the function has a **decreasing** y-value as the x-value increases. This is called a **decreasing function**.
  - A **gradient of zero** means the function is **neither** decreasing or increasing.

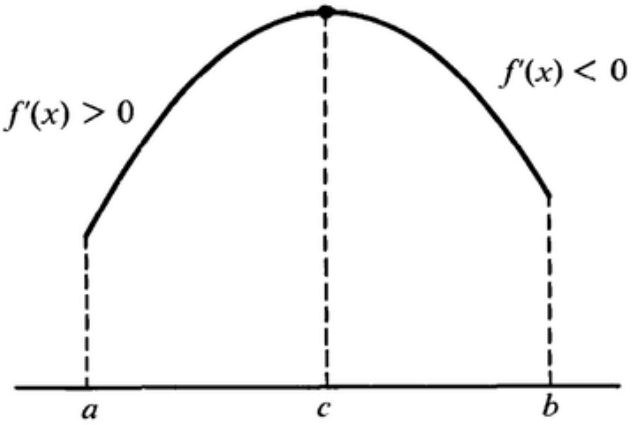


Using the graph above, we can re-write the rate of change as

Gradient function - $f\prime(x)$	Gradient	Function - $f(x)$
$f\prime(x) > 0$	Increasing	Increasing function
$f\prime(x) = 0$	Stationary	Stationary point
$f\prime(x) < 0$	Decreasing	Decreasing function

- It is also important to note that $f\prime(x) > 0$ **does not always imply** $f(x) > 0$, vice versa.

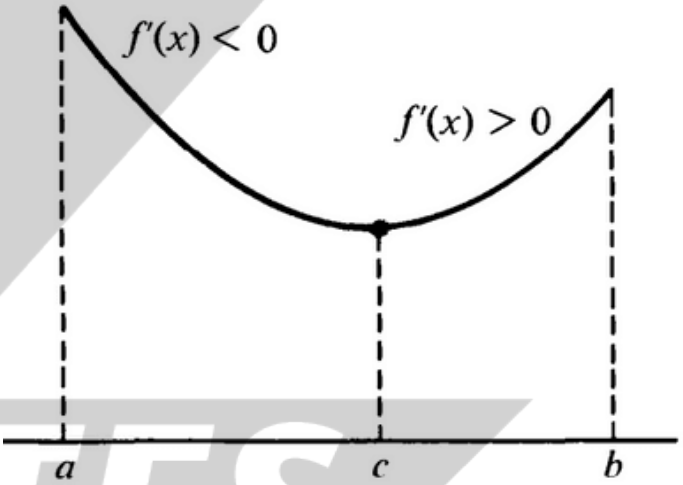
Maximum point



Given a **maxima** with x coordinate c:

Left of maxima	Right of maxima
Increasing gradient, $f\prime(x) > 0$	Decreasing gradient, $f\prime(x) < 0$
Function is **increasing**	Function is **decreasing**

Minimum point



Given a **minima** with x coordinate c:

Left of minima	Right of minima
Decreasing gradient, $f\prime(x) < 0$	Increasing gradient, $f\prime(x) > 0$
Function is **decreasing**	Function is **increasing**

Example 1

A function is defined by

$$f(x) = \frac{5}{1-3x} \text{ for } x \geq 1$$

Determine wether f is an increasing function, decreasing function, or neither.

Answer

First we find the first derivative, or the gradient function, of $f(x)$ using the chain rule.

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$$\frac{d}{dx}\,5(1-3x)^{-1} = -1 \times 5 \times -3 \times (1-3x)^{-2}$$

$$f\prime(x) = 15 \times (1-3x)^{-2}$$

As:

- The 15 being multiplied is positive, it does not change the sign of the function.
- $(1-3x)^{-2}$ is always positive as it is squared, it does not change the sign of the function.

Using this logic, we can deduce that

$$f\prime(x) > 0$$

So $f(x)$ is an increasing function for all real values of x.

Example 2

Find the set of values of x for which the function $f(x) = \frac{(2x-3)^3}{6} - 4x$ is increasing.

Answer

We start by finding the first derivative, or gradient function, of $f(x)$ using the chain rule.

$$\frac{1}{6} \times \frac{d}{dx}\,(2x-3)^3 + \frac{d}{dx} - 4x = \frac{1}{6}\left(3 \times 2 \times (2x-3)^2\right) - 4$$

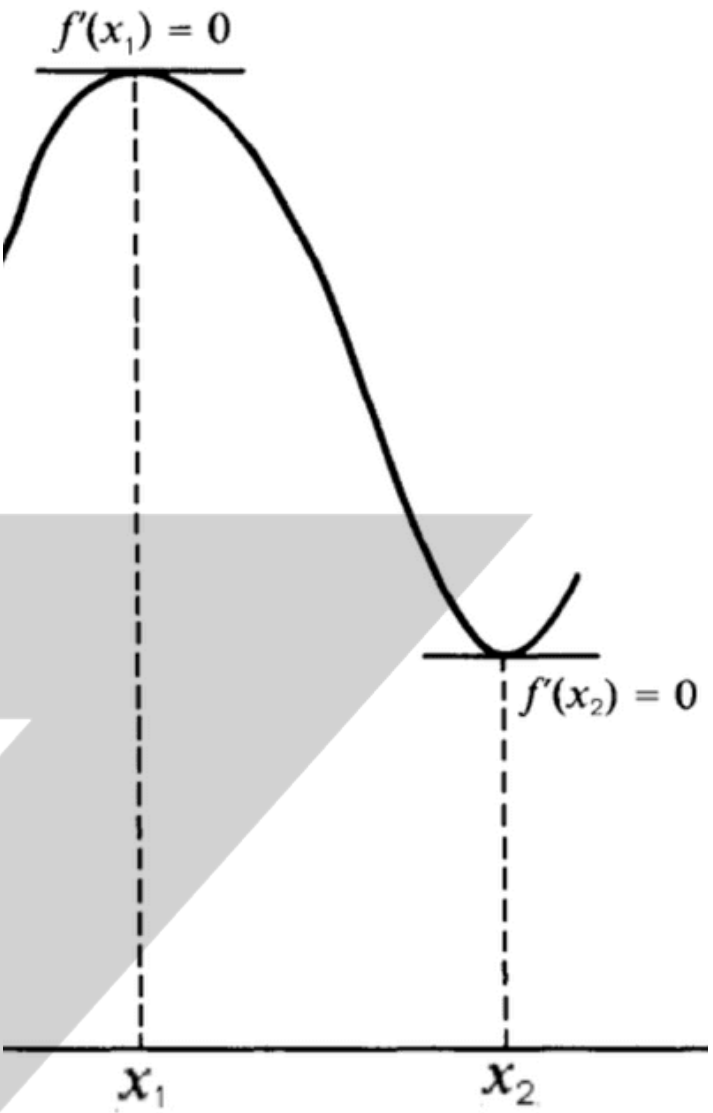
$$f\prime(x) = (2x-3)^2 - 4$$

The function is increasing when $f\prime(x) > 0$

$$(2x-3)^2 - 4 > 0 \implies x > \frac{5}{2} \text{ or } x < \frac{1}{2}$$

1.5. Stationary points

- A stationary, or turning, point is a point on the graph where the gradient is 0.
  - Graphically - The gradient of the tangent, $f\prime(x)$, is horizontal and parallel to the x-axis at a stationary point.



In this case:

- x_1 represents a local **maxima**, or a **maximum point**.
- x_2 represents a local **minima**, or a **minimum point**.

First derivative

- The first derivative helps us find the x coordinate of the stationary point.

To find the **stationary point**:

- Differentiate the function, $f(x)$, to get the first derivative, $f\prime(x)$.
  - $\frac{d}{dx}\,f(x) = f\prime(x)$
- Equate $f\prime(x)$ to 0, as the gradient of the tangent at any stationary point is always 0. This will give you the x coordinate of the stationary point.
  - $f\prime(x) = 0$
- Input the value of the x coordinate, or coordinates in some cases, back into the original equation $f(x)$ to get the y coordinate of the stationary point.

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Second derivative

- The second derivative helps us find the **nature of the stationary point**.
  - It helps us figure out if the stationary point is a **minima** or **maxima**.

To find the second derivative, **differentiate the first derivative**.

$$\frac{d}{dx}\left(\frac{d}{dx}\right) = \frac{d}{dx} f\prime(x) = f\prime\prime(x)$$

It is also denoted by

$$\frac{d^2}{dx^2} \text{ or } \frac{d^2y}{dx^2}$$

Inputting the x coordinate of the stationary point into the second derivative, $f\prime\prime(x)$, will tell us the nature of the stationary point:

Inequality	Nature of stationary point
$\frac{d^2y}{dx^2} > 0$	Minimum point
$\frac{d^2y}{dx^2} < 0$	Maximum point

Example

A curve has equation

$$f(x) = x^2 + \frac{2}{x}$$

Find the coordinates of the stationary point on $f(x)$ and determine its nature.

Answer

To find the coordinate of the stationary point, we find the first derivative and equate it to 0.

$$\frac{d}{dx} x^2 + 2x^{-1} = 2x - 2x^{-2}$$

$$2x - 2x^{-2} = 0 \implies x = 1$$

Inputting the x coordinate back into the original equation, $f(x)$, will give us the y coordinate of the stationary point.

$$f(1) = 1^2 + \frac{2}{1} = 3$$

So the coordinate of the stationary point is

$$(1, 3)$$

To find the nature of this stationary point, we find the second derivative by differentiating the first derivative

$$f\prime\prime(x) = \frac{d}{dx}2x - 2x^{-2} = 2 + 4x^{-3}$$

We can now input the x coordinate of the stationary point into $f\prime\prime(x)$

$$f\prime\prime(1) = 2 + 4(1)^{-3} = 6$$

As $\frac{d^2}{dx^2} > 0$, this stationary point is a **minima**.

1.6. Applications of Differentiation

- Differentiation can be used in modelling situations
  - Such as finding the maximum or minimum volume/area a model can have.
- If the variables in an equation vary with time, or any other variable, their rates of change can be connected using

$$\frac{dy}{dx} = \frac{dt}{dx} \times \frac{dy}{dt}$$

Example

The volume, V, of a spherical balloon is increasing at a constant rate of 50 cm³/s. Find the rate of increase of the radius when the radius is 10 cm.

Answer

We are given $\frac{dV}{dt} = 50$, and are asked to find the value of $\frac{dr}{dt}$.

To solve the question, we first note that the volume of a sphere is given by

$$V = \frac{4}{3}\pi r^3$$

By differentiating the formula for a volume of a sphere, we get

$$\frac{dV}{dr} = 4\pi r^2$$

So at $r = 10$

$$\frac{dV}{dr} = 4\pi(10)^2 = 400\pi$$

Using the connected rates of change formula, or chain rule:

$$\frac{dV}{dr} = \frac{dt}{dr} \times \frac{dV}{dt}$$

We can substitute in our values

$$400\pi = \frac{dt}{dr} \times 50$$

$$\frac{dt}{dr} = \frac{400\pi}{50} = 8\pi$$

To find the value of $\frac{dr}{dt}$ we can take the reciprocal of the result

$$\frac{dt}{dr} = \frac{1}{8\pi}$$

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